

Solutions for CSE303 Homework 5

1. Construct nondeterministic pushdown automata (npda) that accept the following regular languages.
 Note: Observe that all the languages are regular languages, so the solutions are essentially NFA's (or npda's with inactive stack). For all the languages, f is the final state.

(a) $L_1 = L(aaa^*b)$

Solution: The npda $M = (K, \Sigma, \Gamma, \Delta, s, F)$, where $K = \{q_0, q_1, q_2, f\}$, $\Sigma = \{a, b\}$, $F = \{f\}$, and Δ as the set of following rules accepts L_1 .

$$\begin{aligned} &((q_0, a, e), (q_1, e)) \\ &((q_1, a, e), (q_2, e)) \\ &((q_2, a, e), (q_2, e)) \\ &((q_2, b, e), (f, e)) \end{aligned}$$

(b) $L_2 = L(abb^*aba^*)$

Solution: The npda $M = (K, \Sigma, \Gamma, \Delta, s, F)$, where $K = \{r_0, r_1, r_2, r_3, f\}$, $\Sigma = \{a, b\}$, $F = \{f\}$, and Δ as the set of following rules accepts L_2 .

$$\begin{aligned} &((r_0, a, e), (r_1, e)) \\ &((r_1, b, e), (r_2, e)) \\ &((r_2, b, e), (r_2, e)) \\ &((r_2, a, e), (r_3, e)) \\ &((r_3, b, e), (f, e)) \\ &((f, a, e), (f, e)) \end{aligned}$$

(c) the union of L_1 and L_2

Solution: The npda $M = (K, \Sigma, \Gamma, \Delta, s, F)$, where $K = \{s, q_0, q_1, q_2, r_0, r_1, r_2, r_3, f\}$, $\Sigma = \{a, b\}$, $F = \{f\}$, and Δ as the set of following rules accepts $L_1 \cup L_2$

$$\begin{aligned} &((s, e, e), (q_0, e)) \\ &((s, e, e), (r_0, e)) \end{aligned}$$

where q_0 and r_0 are as defined above.

2. Construct npda's that accept the following languages on $\Sigma = \{a, b, c\}$.

(a) $L = \{a^n b^{2n} : n \geq 0\}$

Solution: The npda $M = (K, \Sigma, \Gamma, \Delta, s, F)$, where $K = \{s, f\}$, $\Sigma = \{a, b\}$, $F = \{f\}$, and Δ as the set of following rules accepts L .

$$\begin{aligned} &((s, e, e), (f, e)) \\ &((s, a, e), (s, aa)) \\ &((s, b, a), (f, e)) \\ &((f, b, a), (f, e)) \end{aligned}$$

(b) $L = \{a^n b^m c^{n+m} : n \geq 0, m \geq 0\}$

Solution: The npda $M = (K, \Sigma, \Gamma, \Delta, s, F)$, where $K = \{s_0, s_1, f\}$, $\Sigma = \{a, b\}$, $F = \{s_1\}$, and Δ as the set of following rules accepts L .

$$\begin{aligned} &((s_0, e, e), (f, e)) \\ &((s_0, a, e), (s_0, a)) \\ &((s_0, c, a), (f, e)) \\ &((s_0, b, e), (s_1, b)) \\ &((s_1, b, e), (s_1, b)) \\ &((s_1, c, b), (f, e)) \\ &((f, c, b), (f, e)) \\ &((f, c, a), (f, e)) \end{aligned}$$

3. Find an npda on $\Sigma = \{a, b, c\}$ that accepts the language $L = \{w_1cw_2 : w_1, w_2 \in \{a, b\}^*, w_1 \neq w_2^R\}$

Solution: Idea: Observe that if $w_1 = w_2^R$, then the npda should not accept the input. But $w_1 = w_2^R$ is equivalent to $w_1^R = w_2$. Hence, we first, push w_1 onto stack, then while matching w_2 , if a mismatch is found (then $w_1^R \neq w_2$), consume all the input and empty the stack then go to final state; otherwise (then $w_1^R = w_2$ which means $w_1 = w_2^R$), the input is not accepted. Thus the npda $M = (K, \Sigma, \Gamma, \Delta, s, F)$, where $K = \{s_0, s_1, d, f\}$, $\Sigma = \{a, b\}$, $F = \{f\}$, and Δ as the set of following rules accepts L .

$$\begin{aligned} &((s_0, a, e), (s_0, a)) \\ &((s_0, b, e), (s_0, b)) \\ &((s_0, c, e), (s_1, e)) \\ &((s_1, a, a), (s_1, e)) \\ &((s_1, b, b), (s_1, e)) \\ &((s_1, a, b), (d, e)) \\ &((s_1, b, a), (d, e)) \\ &((d, e, a), (d, e)) \\ &((d, e, b), (d, e)) \\ &((d, a, e), (d, e)) \\ &((d, b, e), (d, e)) \\ &((d, e, e), (f, e)) \end{aligned}$$

4. Construct an npda corresponding to the grammar

$$\begin{aligned} S &\rightarrow aABB \mid aAA, \\ A &\rightarrow aBB \mid a, \\ B &\rightarrow bBB \mid A \end{aligned}$$

Solution: The npda $M = (K, \Sigma, \Gamma, \Delta, s, F)$, where $K = \{s_0, s_1\}$, $\Sigma = \{a, b\}$, $F = \{s_1\}$, and Δ as the set of following rules implements the grammar above.

$$\begin{aligned} &((s_0, e, e), (s_1, S)) \\ &((s_1, a, S), (s_1, ABB)) \\ &((s_1, a, S), (s_1, AA)) \\ &((s_1, a, A), (s_1, BB)) \\ &((s_1, a, A), (s_1, e)) \\ &((s_1, b, B), (s_1, BB)) \\ &((s_1, a, B), (s_1, BB)) \\ &((s_1, a, B), (s_1, e)) \end{aligned}$$

Another solution:

$$\begin{aligned} &((s_0, e, e), (s_1, S)) \\ &((s_1, e, S), (s_1, aABB)) \\ &((s_1, e, S), (s_1, aAA)) \\ &((s_1, e, A), (s_1, aBB)) \\ &((s_1, e, B), (s_1, bBB)) \\ &((s_1, e, B), (s_1, A)) \\ &((s_1, a, a), (s_1, e)) \\ &((s_1, b, b), (s_1, e)) \end{aligned}$$

5. Show that the language $L = \{ww : w \in \{a, b\}^*\}$ is not context-free.

Solution: Consider the string $a^m b^m a^m b^m$. Now, let v and y contain only the first a 's and let $v = a^p$ and $y = a^q$. Then consider $uxz = a^k b^m a^m b^m$, $k < m$, which is not in L . For other choices of v and y , we can make similar arguments (since each symbol occurs exactly m times). Thus, L is not context-free.

6. Show that the language $L = \{a^{n!} : n \geq 0\}$ is not context-free.

Solution: In this case, this is same as showing that L is not regular (since the language consists entirely of alphabet over single symbol). Let $w = uvxyz$ such that $uv = \epsilon$. Then we need to show that $w = xy^iz$, for $i = 0, 1, 2, \dots$. Let $w = a^{m!}$. Let $|y| = k \leq m$. Then xz has length $m! - k$. This string is in L only if there exists a j such that $m! - k = j!$. But this is impossible, since for $m > 2$ and $k \leq m$, we have $m! - k > (m-1)!$. Therefore L is not context-free (or regular, either).

7. Construct Turing machines that will accept the following languages on $\{a, b\}$.

- (a) $L = \{w : |w| \text{ is even}\}$

Solution: Here we keep checking off two input symbols and if ultimately we encounter end of input in the start state, we accept the input.

The Turing machine $M = (K, \Sigma, \delta, s, H)$, where $K = \{q_0, q_1, h\}$, $\Sigma = \{a, b, \sqcup, \triangleright\}$, $s = q_0$, $H = \{h\}$ and δ as follows accepts L .

q	σ	$\delta(q, \sigma)$
q_0	a	$(q_1, \rightarrow)$
q_0	b	$(q_1, \rightarrow)$
q_1	a	$(q_0, \rightarrow)$
q_1	b	$(q_0, \rightarrow)$
q_0	$\sqcup$	$(h, \sqcup)$

- (b) $L = \{w : n_a(w) = n_b(w)\}$

where $n_a(w)$ is number of a 's in w and $n_b(w)$ is number of b 's in w .

Solution: First find leftmost a or b . If a (or b) is found, then replace it with x and find a matching b (or a), replace it with x and then go back to the left end of the tape (so as to find next leftmost a or b). Repeat.

The Turing machine $M = (K, \Sigma, \delta, s, H)$, where $K = \{q_0, q_1, q_2, q_3, h\}$, $\Sigma = \{a, b, \sqcup, \triangleright\}$, $s = q_0$, $H = \{h\}$ and δ as given below accepts L .

q	σ	$\delta(q, \sigma)$
q_0	x	$(q_0, \rightarrow)$
q_0	a	(q_1, x)
q_0	b	(q_2, x)
q_1	x	$(q_1, \rightarrow)$
q_1	a	$(q_1, \rightarrow)$
q_1	b	(q_3, x)
q_2	x	$(q_2, \rightarrow)$
q_2	a	(q_3, x)
q_2	b	$(q_2, \rightarrow)$
q_3	x	$(q_3, \leftarrow)$
q_3	a	$(q_3, \leftarrow)$
q_3	b	$(q_3, \leftarrow)$
q_3	$\triangleright$	$(q_0, \rightarrow)$
q_0	$\sqcup$	$(h, \sqcup)$